

ASSESSMENT GUIDE

Unit Code: ITEC613 Machine Learning with Scikit-Learn 2025 Study

Period: S1 2025

Assessment number (3)

Assessment Artefact: Code and Presentation

Weighting [40%]

Why this assessment?

- Opportunity to apply theory into practice
- Exposure to real-life scenario
- Enhance the understanding of theoretical concepts introduced in workshops – the definition of each problem, the logic of the algorithms, the meaning of their parameters and hyperparameters, the cons and pros of them.
- Build up students' coding and problem-solving skills with coding practice.
- Build up students' documentation skills with report output.
- Build up students' teamwork ability
- The feedback from this assessment will help students to be ready to correct any conceptual misunderstanding and apply in more scenarios when taking roles e.g. data engineer in the job markets.

What are the types of employability skills that I will acquire upon completion of this assessment?

Skill Type	
Developed critical and analytical thinking	☒
Developed ability to solve complex problems	☒
Developed ability to work effectively with others	☒
Developed confidence to learn independently	☒
Developed written communication skills	☒
Developed spoken communication skills	☒
Developed knowledge in the field study	☒
Developed work-related knowledge and skills	☒

Assessment Overview:

Purpose, as written in the EUO

Due date:	09/06/2025, 5pm, on Monday of Week 14
Weighting:	40%

Length and/or format:	Document in the format of Jupyter notebook including Python source codes, Word document and recorded presentation – about 5 minutes
Learning outcomes assessed	LO3, LO4
Graduate attributes assessed	GA2, GA5, GA8
How to submit:	Via CANVAS
Assessment criteria:	Rubric: see end of document

Context

As a senior Data Engineer, your role extends beyond applying machine learning algorithms; it includes optimizing models, ensuring data privacy, and making strategic decisions based on data insights. This assessment is designed to deepen your expertise in machine learning through complex real-world dataset applications.

Instructions

This assignment requires applying advanced classification algorithms to the MNIST dataset and conducting an in-depth regression analysis on the California housing dataset. Each task must demonstrate not only technical proficiency but also strategic thinking and ethical considerations.

Tasks

1. Advanced Dataset Preparation

- **Split the MNIST dataset** into training and testing sets using scikit-learn functions or your own custom function. Include detailed comments to explain your process.
- **Conditions:**
 - Use your first name for the training set and your last name for the testing set variable names. For instance, If your name is john doe, use **john_train** and **doe_test** as your variable names.
 - Use the last two digits of your student ID as the **random_state** for any function that requires it. For instance, the value for the random_state = **[Last two digits of your student ID]**
- Utilize advanced preprocessing techniques to enhance model performance, such as feature scaling and dimensionality reduction where appropriate.

2. Advanced kNN Classifier

- Set **k=10** and utilize the kNN classifier from scikit-learn, providing a detailed explanation of the function parameters and their implications.
- Evaluate the model using advanced metrics. Discuss the rationale behind choosing specific metrics and their implications on the model evaluation.
- Conduct an in-depth analysis of varying **k** values, including a statistical test to determine if changes in performance are significant.

3. SVM Classifier with Parameter Optimization

- Apply an SVM classifier using both linear and non-linear kernels. Experiment with feature engineering techniques to improve model accuracy.
- Discuss the kernel trick and its impact on the computational complexity and performance of the model.

4. **Neural Network Classification and Ethical Implications**

- Implement a deep neural network on the MNIST dataset using TensorFlow or PyTorch.
- Explore and justify the choice of network architecture, layers, and hyperparameters.
- Explore and discuss the ethical implications of model predictions, particularly focusing on issues like fairness, accountability, and transparency in AI.

5. **Comprehensive Model Analysis**

- Compare all classifiers implemented in previous tasks.
- Perform a statistical analysis of their performance metrics (accuracy, precision, recall, F1 score) to determine the best model.
- Discuss each model's suitability for various real-world applications and suggest practical improvements or alternative approaches.

6. **Advanced Regression Analysis**

- Import the california_housing dataset from the sklearn.datasets
- Apply multiple regression models (Linear, Polynomial, and Ridge/Lasso Regression) to the California housing dataset. Discuss the benefits and limitations of each under various data conditions.
- Evaluate model performance with both numerical and graphical analysis, looking at error metrics and residual plots.
- Investigate the impact of different polynomial degrees on performance and overfitting.

7. **Recorded Presentation**

- Prepare a detailed presentation summarizing your methodology, findings, model comparisons, and strategic insights derived from the analysis.
- Use professional visualization tools to support your findings and provide a clear narrative that could influence business or technical decisions.
- Prepare and record a detailed 5-minute presentation that summarizes your methodologies, findings, and insights from all tasks. Ensure clarity and depth in your explanation.

Structure

Prepare a Jupyter notebook and video recordings for this assignment. The structure of the Jupyter notebook should alternate texts and python codes and cover topics listed the in specific tasks above. The video recording is used to confirm academic integrity. All the codes in the Jupyter notebook needs to be explained in the video.

How do I submit?

Step-by-Step Submission Guide:

1. Prepare Your Submission:

- Ensure your Jupyter notebook (.ipynb) is complete with all required tasks, including the implementation and discussion points for each machine learning model.
- Make sure to use variable names that incorporate your first and last names to ensure academic integrity. For example, if your name is John Doe, use **john_train** for your training dataset and **doe_test** for your testing dataset.
- Include detailed comments within your Jupyter Notebook and documentation in another word file to explain your code and your findings.

2. Access CANVAS:

- Go to the CANVAS course page for "Machine Learning with Scikit-Learn"
- Navigate to the "Assignments" section and find the link for "Assessment Task 3."

3. Upload Files:

- Upload your Jupyter notebook file (.ipynb) and/or the pdf version containing all the code and analysis for the tasks.
- Upload all the other files such as the detailed reports as Word documents or PDFs, and also your recorded presentation.

4. Submit:

- Review all files to ensure they are complete and accurate.
- Click "Submit" to complete your submission process.

Important Notes:

- **Plagiarism Check:** Your submission will be checked for plagiarism using Turnitin to ensure that the work is original and not copied from other students. It is crucial that you write your own code and provide personal insights in the analysis to maintain academic integrity.
- **Deadline:** Make sure to submit your assessment by the due date: **09/06/2025 by 5pm**. Late submissions may be subject to penalties as per course policy.

Ensuring Academic Integrity:

- Use unique variable names linked to your name as part of the code to demonstrate personal effort and reduce the likelihood of code sharing.
- Explain your code decisions and model choices thoroughly in your Jupyter notebook. This not only helps in demonstrating your understanding but also ensures that your submission is distinct and tailored to the specific tasks of this assessment.

Submission checklist

I have formatted my report as per the specifications	☐
I have checked my Turnitin report and taken appropriate actions to ensure that the submission satisfies academic integrity	☐
I have actioned feedback advice provided to me from labs feedback and assignment 2(if applicable)	☐
I have submitted my work before the due date/time	☐
I have submitted feed forward template along with my assignment submission	☐

Feed Forward Template (example)

A template for students to use and act on feedback and provide recommendations for improvement.

Note

This is a task for any instance of follow-on assignment (assessment 2 and 3). This must be submitted as the first page of the follow-on assignment (assessment 2 and 3) to ensure you acted on the feedback provided to you in the previous assignment (this is not counted as part of the assessment word count).

How did you act on the feedback?

Feedback is an important component of learning. Please consider the feedback you received in your last assignment and provide a response on how you acted on, or intend to act upon, that feedback, and how it has informed the current assignment task. Submit this sheet along with your assignment.

Questions	Your learning from the previous assignment feedback
How have you acted on the feedback from previous assignment to improve your work in this assignment?	(e.g. based on my previous feedback, I made sure that I supported my discussion, position, ideas, concepts with peer reviewed journal references in this assignment)
What is your expectation around the type of feedback that enhances your learning?	(e.g. I want to know where I made a mistake and how I can correct them and not make the same mistake again i.e. I want specific feedback that will help me to improve my learning and performance in the next assignment)
Did you have any difficulty understanding or acting on previous feedback? Please be as specific as possible so that you can gain further feedback/clarify anything you do not understand in the feedback	(e.g. feedback provided in my previous assignment was very generic I did not know how to improve my work. So, I would like the teacher to explain more on xxxx aspects of the feedback or I would like an opportunity to have a dialogue to understand the feedback)

Some Helpful Websites and Resources

Add in a couple of places to go for more info

Model Evaluation: https://scikit-learn.org/stable/auto_examples/model_selection/plot_roc.html

Cross-validation: https://scikit-learn.org/stable/modules/cross_validation.html

Grid search: https://scikit-learn.org/stable/modules/grid_search.html

Who can help me?

Studiosity

Academic skills Unit (ASU)

Places –NLiC Mahmoud Bekhit (Mahmoud.bekhit@acu.edu.au)

I'm having problems

Special Consideration: This [form](#) is used by students to apply for Special Consideration for assessable work in studies at Australian Catholic University. Approval of such applications will only be granted to students who are legitimately disadvantaged in their assessment due to exceptional and unforeseen circumstances beyond their control.

Referencing

All referencing should be in [ACU Harvard style](#); however if you are coming from another faculty, you may choose to use your usual referencing style. If this is the case you must indicate at the top of your reference list what referencing style you are using (e.g. APA, MLA, Chicago, etc).

Please ensure your assignment makes use of in-text citations and a reference list. Missing citations or references is equivalent to plagiarism.

Criteria

The full criteria is compiled in a rubric, which can be found on the following page/s.

Rubric for [ITEC613 AT3, 40%]

Relevant LO/GAs	Criterion (related to a single GA from the related LO – one GA per criterion)	Does not meet expectations	Meets expectations	Exceeds expectations		
		NN	PA	CR	DI	HD
GA5 LO3 Weight=20 marks TL=3 Learning stage = I and D	Demonstrate correct understanding of the concepts of classification, kNN, SVM, NN, Regression training-testing splitting, evaluation	Fail to adequately demonstrate correct understanding of the concepts of classification, kNN, SVM, NN, Regression, training-testing splitting, evaluation (0 – 9.8)	Adequately demonstrate correct understanding of the concepts of classification, kNN, SVM, NN, Regression, training-testing splitting, evaluation (9.8 – 12.8)	Credibly demonstrate correct understanding of the concepts of classification, kNN, SVM, NN, Regression, training-testing splitting, evaluation (12.8– 14.8)	Distinctively demonstrate correct understanding of the concepts of classification, kNN, SVM, NN, Regression, training-testing splitting, evaluation (14.8– 16.8)	Highly distinctively demonstrate correct understanding of the concepts of classification, kNN, SVM, NN, Regression, training-testing splitting, evaluation (16.8– 20)
GA4 LO3 Weight=15 marks TL=3 Learning stage = I and D	Demonstrate python programming skills by experimenting with related functions in scikit-learn package for classification and evaluation as well as model selection.	Fail to adequately demonstrate python programming skills by experimenting with related functions in scikit-learn package for classification and evaluation as well as model selection. (0 – 7.35)	Adequately demonstrate python programming skills by experimenting with related functions in scikit-learn package for classification and evaluation as well as model selection. (7.5 – 9.6)	Credibly demonstrate python programming skills by experimenting with related functions in scikit-learn package for classification and evaluation as well as model selection. (9.75 – 11.1)	Distinctively demonstrate python programming skills by experimenting with related functions in scikit-learn package for classification and evaluation as well as model selection. (11.25 – 12.6)	Highly distinctively demonstrate python programming skills by experimenting with related functions in scikit-learn package for classification and evaluation as well as model selection. (12.75 – 15)
GA2 LO4 Weight=5 marks TL=3 Learning stage = I and D	Demonstrate awareness of data ethics occurred in the data manipulation process	Fail to adequately demonstrate awareness of data ethics occurred in the data manipulation process, i.e., no data is explored (0 – 2.45)	Adequately demonstrate awareness of data ethics occurred in the data manipulation process, i.e., part of the data is explored via one task (2.45 – 3.2)	Credibly demonstrate aware of data ethics occurred in the data manipulation process, i.e., part of the data is explored via two tasks (3.2 – 3.7)	Distinctively demonstrate awareness of data ethics occurred in the data manipulation process, i.e., part of the data is explored via most of the tasks (3.7 – 4.2)	Highly distinctively demonstrate awareness of data ethics occurred in the data manipulation process, i.e., full data is explored via all tasks . (4.2 – 5)

